



Loop a Pattern in Code

Quick facilitation notes & answer key

Goal

Students see a for-loop repeat a multi-move PATTERN (across-then-up) to build a staircase, and count moves and total height by multiplying per-step amounts by the repeat count.

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

Python Turtle projected (or a web playground); the printed worksheet + a pencil per student. Graph paper helps draw the staircase.

Run it (about 30 min)

1. Walk a staircase as a class — across, up, across, up — name the repeating step.
2. Run the `range(3)` staircase; trace one step, then the repeat.
3. Students do Exercises 1–4 (Ex 4 multiplies the rise), solo or in pairs.
4. Share answers; land the big idea: a loop can repeat a pattern of moves.

Answer key (checked by the run-gate)

Ex 1 — 3 steps; 6 forward moves (2 per step $\times$ 3).

Ex 2 — 5 steps; 10 forward moves.

Ex 3 — `range(4)`.

Ex 4 (challenge) — 120 high (4 steps $\times$ 30 up).

Watch for & differentiate

Common slip: counting one forward per step instead of two.

Support: highlight the two forward lines inside the loop before counting.

Extend: ask the height of a 6-step staircase (180).

Success looks like

The student counts steps and total forwards (Ex 1–2), writes the range for a 4-step staircase (Ex 3), and multiplies the rise to find the total height (Ex 4).